

Week 2: Understanding Embeddings Like an Engineer (Beginner)

1. The Problem: One-Hot Encoding

We have warehouse words.

Word	One-Hot Vector (7 dims)						
Inventory	1	0	0	0	0	0	0
Order	0	1	0	0	0	0	0
Shipment	0	0	1	0	0	0	0
Receive	0	0	0	1	0	0	0
Restock	0	0	0	0	1	0	0
Forecast	0	0	0	0	0	1	0
Scenario	0	0	0	0	0	0	1

Like giving each word a unique ID
Inventory → 0 Order → 1 Shipment → 2
Receive → 3 Restock → 4 ...

2. What's Wrong With One-Hot?

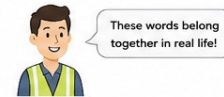
To the computer, all words are equally unrelated.



One-hot vectors cannot represent relationships between words.

3. But Humans Know Better

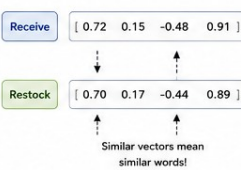
A warehouse worker naturally connects these words.



The neural network should learn these relationships too.

4. The Idea Behind Embeddings

Instead of IDs, give each word a learned profile (a vector).



Like GPS coordinates:
Numbers alone mean nothing, but nearby points are similar.

5. Where Are These Numbers Stored?

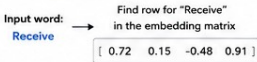
Inside the neural network in a table called the **Embedding Matrix**.

Word	Dim 1	Dim 2	Dim 3	Dim 4	...
Inventory	0.11	0.82	-0.21	0.44	...
Order	0.91	0.18	-0.54	0.03	...
Shipment	0.61	0.52	-0.11	0.82	...
Receive	0.72	0.15	-0.48	0.91	...
Restock	0.69	0.17	-0.46	0.88	...
...	...	...	...	...	...

Each row = one word's embedding vector.

6. How Does the Network Use It?

Instead of using a one-hot vector, the model does an **Embedding Lookup**.



It's simply retrieving one row from a table. No magic!

7. Where Do These Numbers Come From?

At the beginning, the numbers are random and have no meaning.



Random initialization
The model hasn't learned anything yet.

8. How Does Learning Happen?



The model makes predictions → sometimes wrong → error
Backpropagation computes the error → Gradient descent updates the numbers.

The embedding table is updated too, just like all other parameters!

9. Imagine Updating a Spreadsheet

The embedding matrix (a small example)

Before training					After one update					After thousands of updates				
Word	X	Y	Z		Word	X	Y	Z		Word	X	Y	Z	
Receive	0.10	-0.80	0.40		Receive	0.12	-0.76	0.42		Receive	0.71	0.82	0.14	
Restock	-0.30	0.50	-0.60		Restock	-0.29	0.48	-0.57		Restock	0.69	0.80	0.18	

Rows become similar automatically.
The model learned it from data!

10. Why Do Similar Words Become Similar?

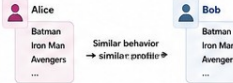
Because they appear in similar situations. The model nudges their vectors in similar directions again and again.



Whenever "Receive" helps predict the same kinds of words as "Restock", their vectors move closer together over thousands of examples.

11. Real-World Analogy (Netflix)

People who watch similar movies get similar profiles.



Embeddings work the same way!
Words used in similar contexts get similar vectors.

12. The Big Picture (End-to-End)

